

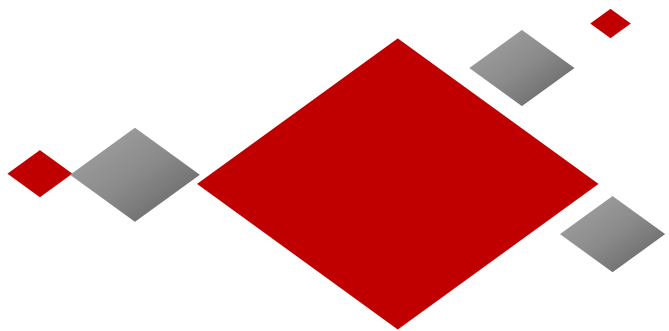


大连理工大学
DALIAN UNIVERSITY OF TECHNOLOGY

Dalian University of Technology Undergraduate Diploma thesis

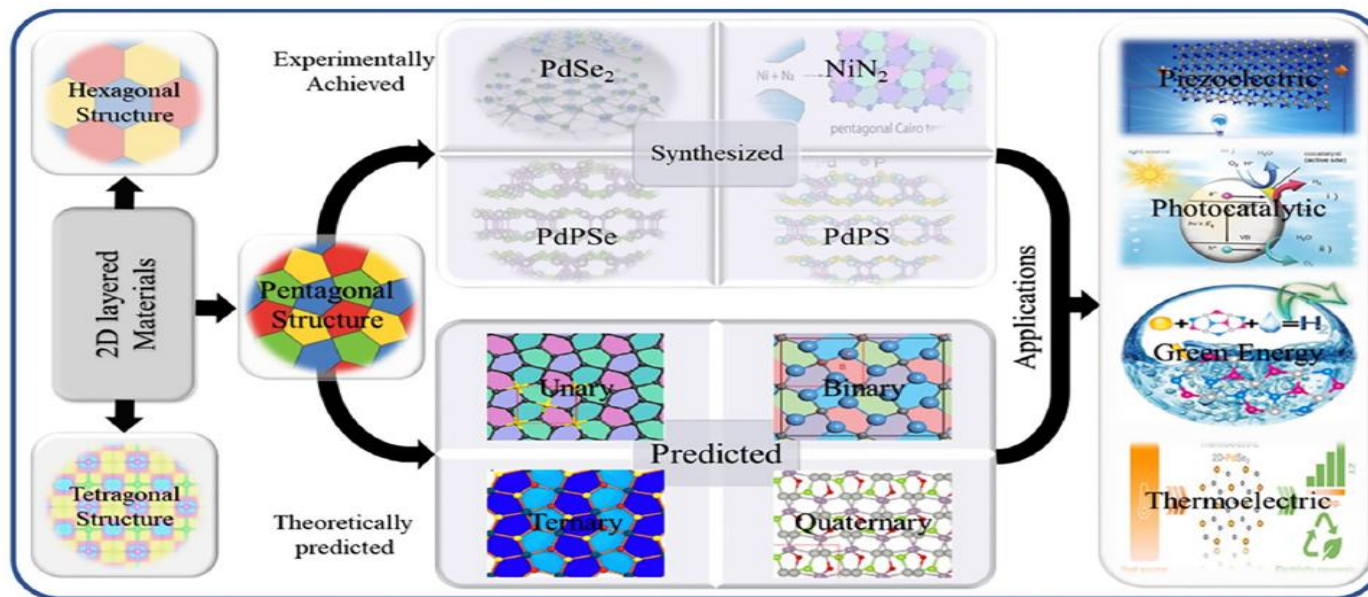
Effect of vdW correction on the electronic properties of pentagonal PdSe₂

Ruizhe Xu | Foundation of Mathematical Science
2026.5.23

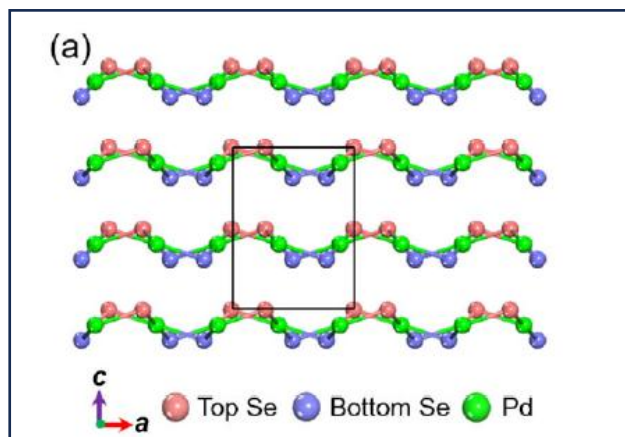


1. Introduction

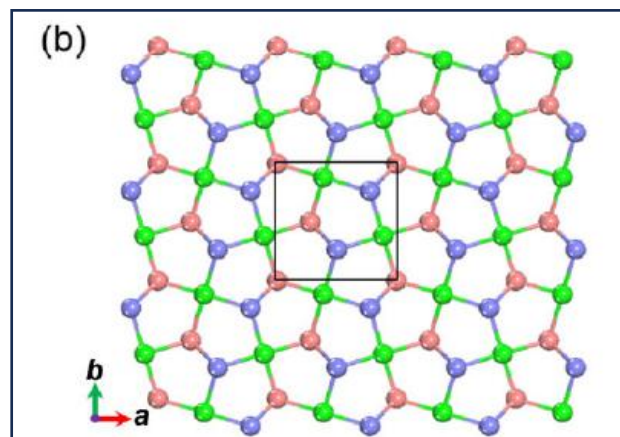




Evolutionary Logic and Applications of pentagonal 2D Materials^[2]



(a) Side view of PdSe₂^[12]



(b) Top view of PdSe₂^[12]

2D Material Evolution

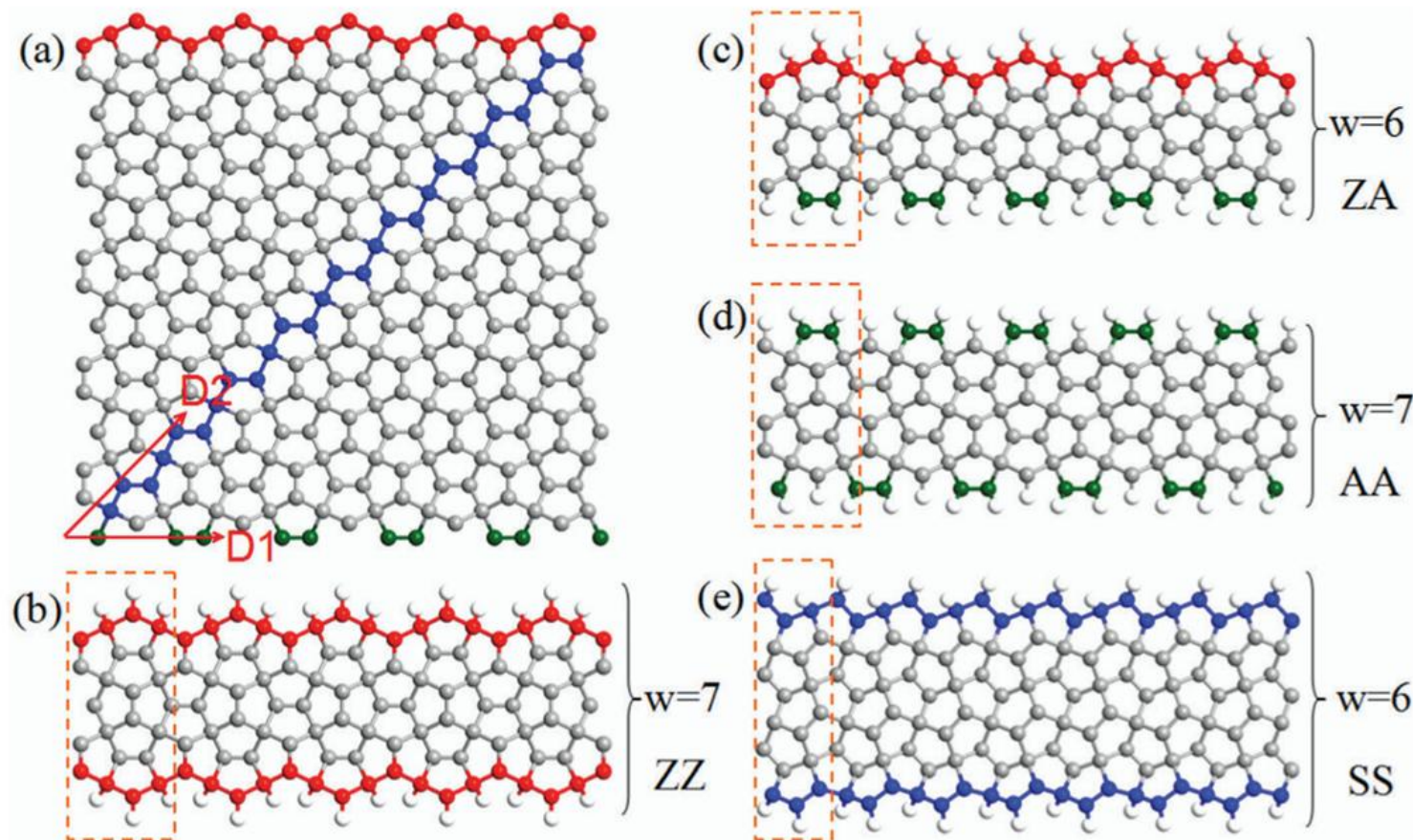
- i. Graphene
- ii. Transition metal dichalcogenides
- iii. Pentagonal 2D materials

Key Properties

- i. Electronic: tunable gap
- ii. Mechanical: Flexibility
- iii. Chemical functionalization

Intrinsic flexibility → quantum-mechanically coupled to the specific atomic-scale structural topology

Understanding the specific characteristics of PdSe_2 is of great significance for regulating its electronic properties.



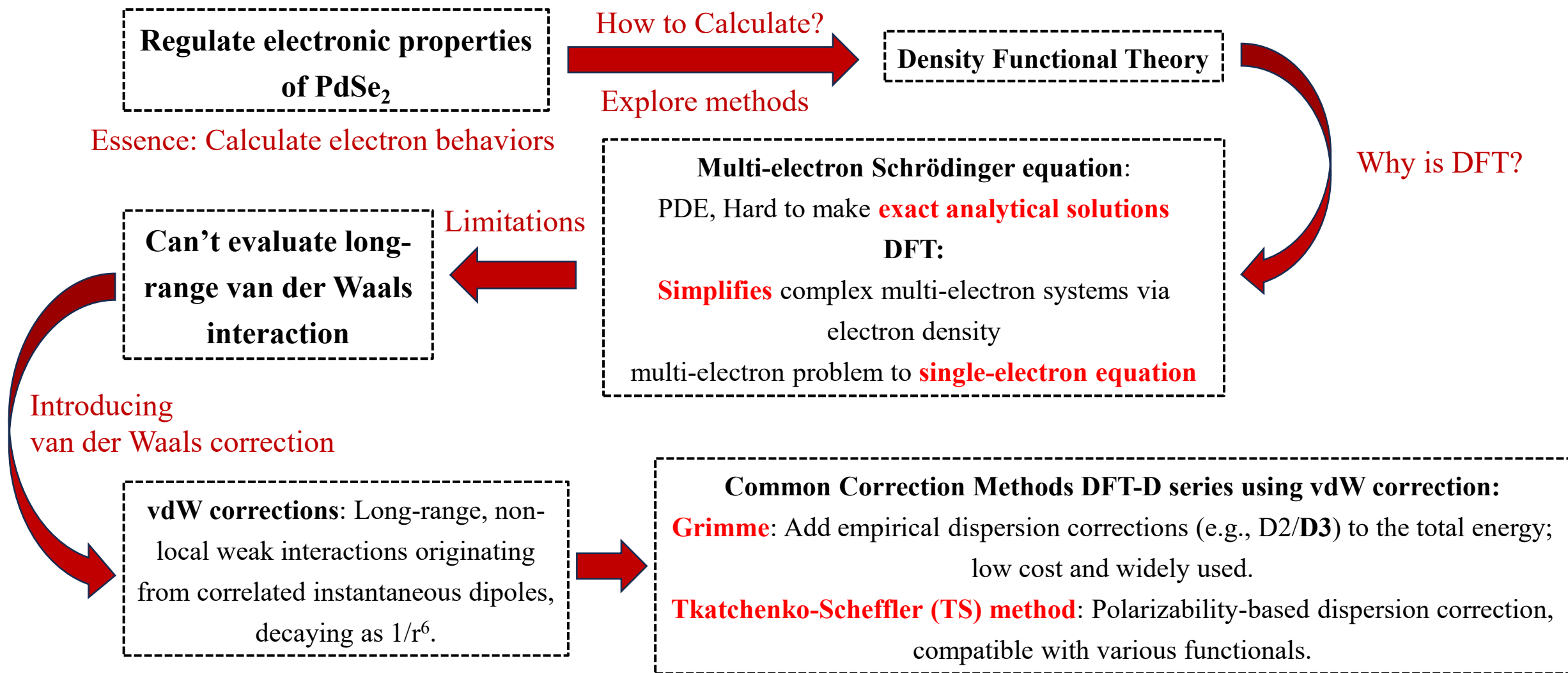
The four edge types [(b)Zigzag-Zigzag, (c)Zigzag-Armchair, (d)Armchair-Armchair, (e)Sawtooth-Sawtooth], formed by different cutting methods(a).^[7]

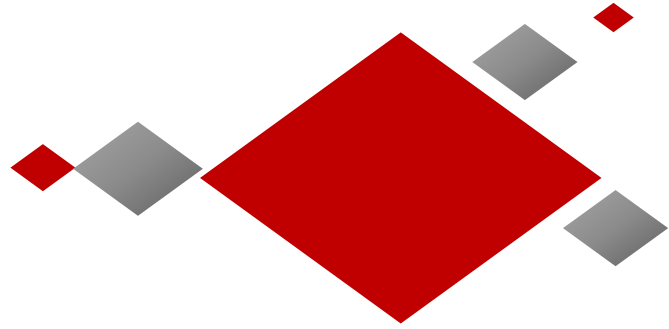
- i. Puckered pentagonal lattice
- ii. Excellent air stability
- iii. Tunable bandgap from 0 eV to 1.3 eV

Advantages of PdSe_2

- i. Low lattice thermal conductivity
- ii. Anisotropic thermoelectric
- iii. Optical and mechanical properties
- iv. Near-zero Poisson's ratio

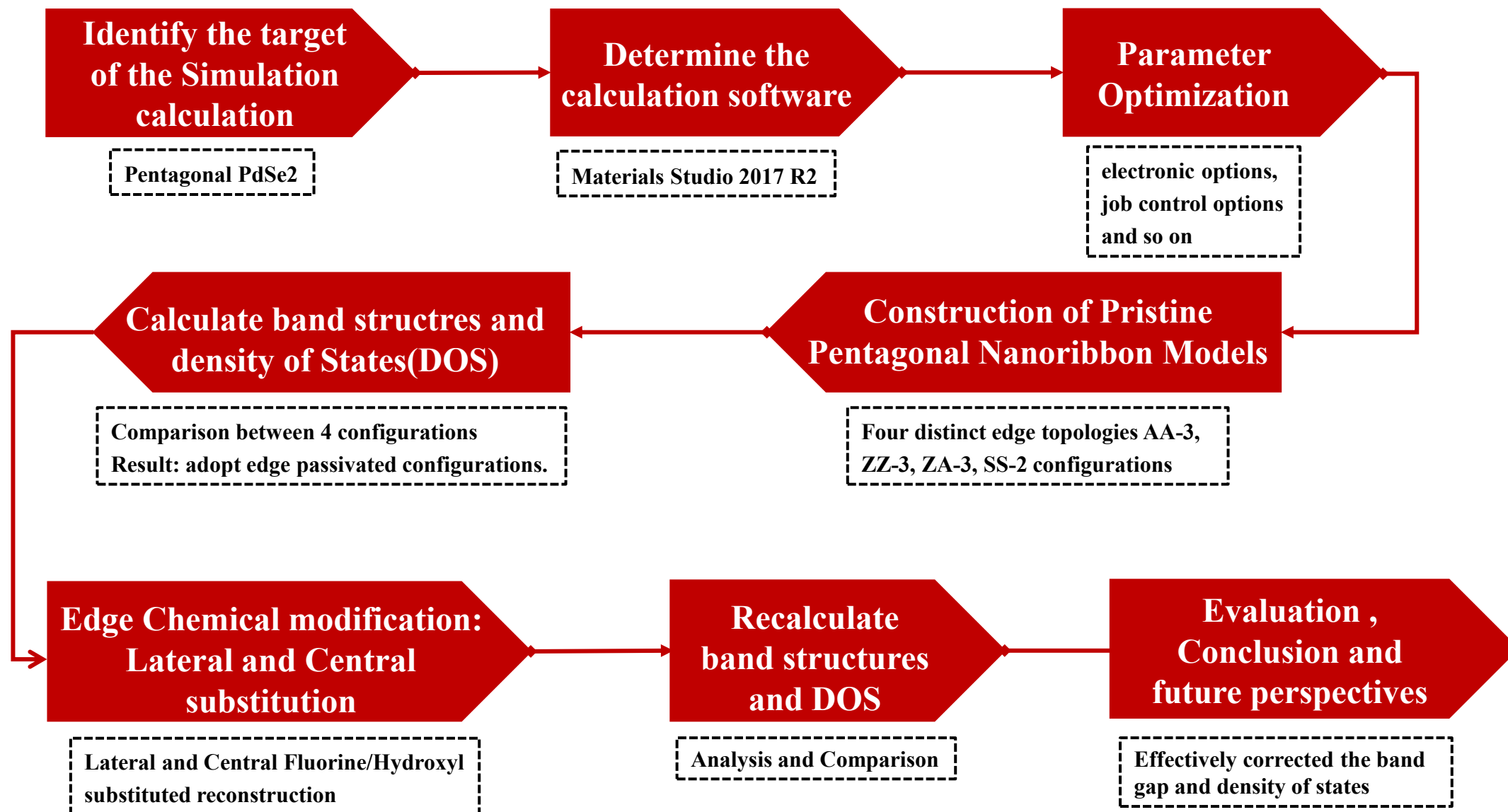
Key Properties of PdSe_2





2. Research plan and Procedure





Software & Basis

All calculations performed via
CASTEP in Materials Studio 2017 R2

Bloch theorem + plane-wave basis set,

OTFG ultrasoft pseudopotential,

cutoff energy = 500 eV

Ultra-fine Convergence Criteria

Energy: 5.0×10^{-6} eV/atom;

Force: 0.01 eV/Å; **Stress:** 0.02 GPa;

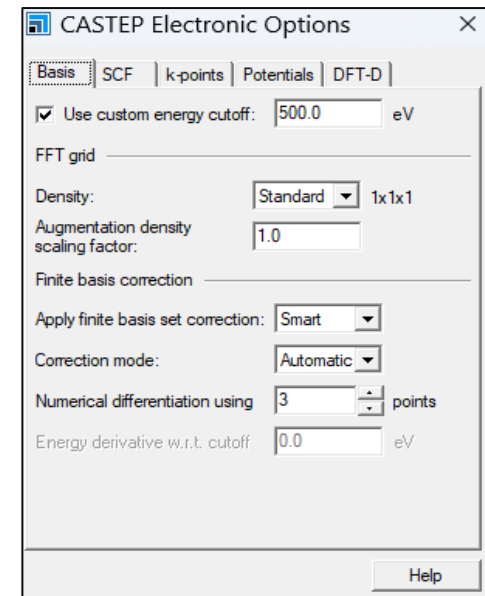
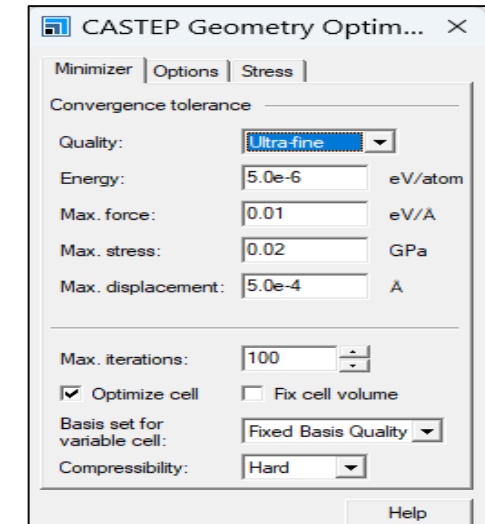
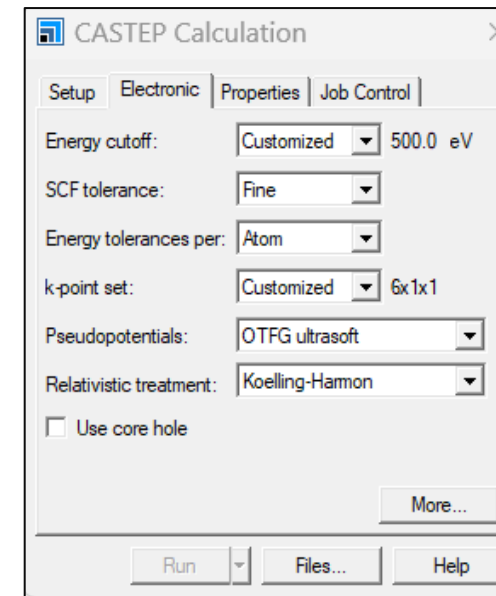
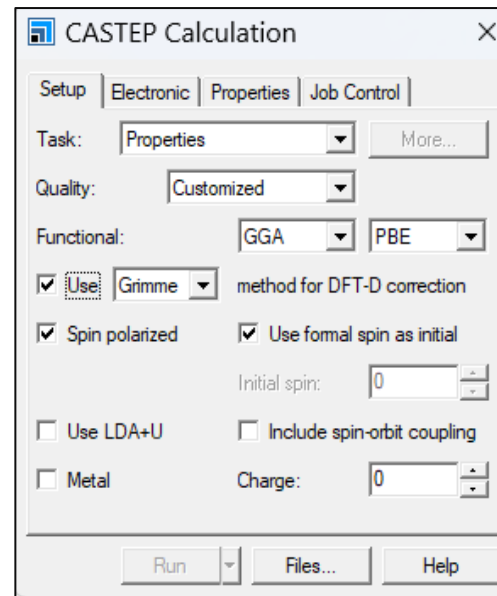
Displacement: 5.0×10^{-4} Å.

k-point mesh Monkhorst-Pack grid

with 0.05 Å^{-1} spacing.

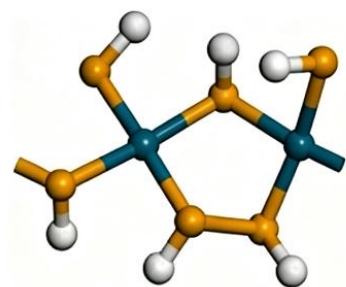
vdW parameters TS correction:

$s_R=0.94$, $d=20.0$.

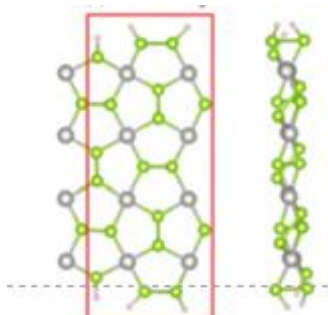


Parameters Setting and Optimizations on Materials Studio

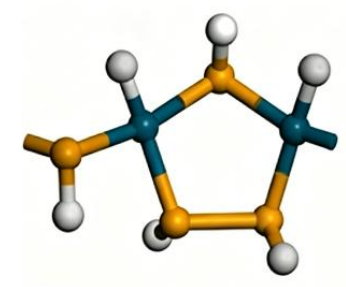
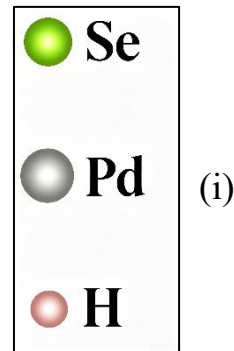
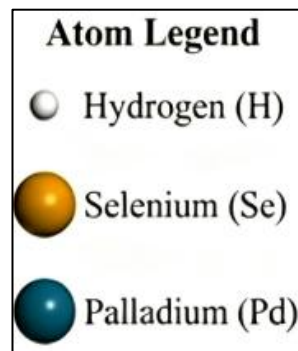
PdSe₂ edge configurations



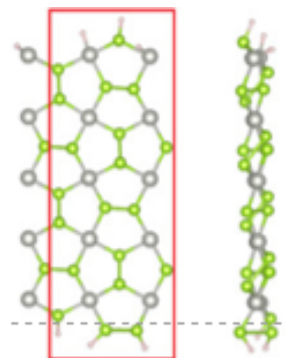
(a)



(e)

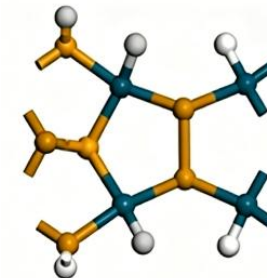


(c)

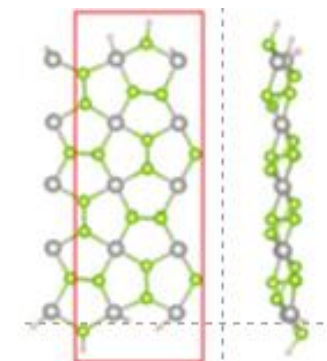


(g)

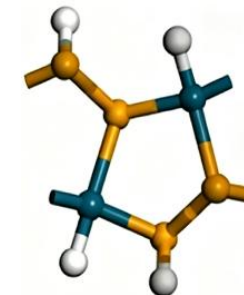
Why choose AA-3, ZZ-3, ZA-3, SS-2?
Narrow ribbons exhibit
Stronger quantum confinement,
Clearer DOS peaks,
More pronounced edge passivation effects
on electronic structure.



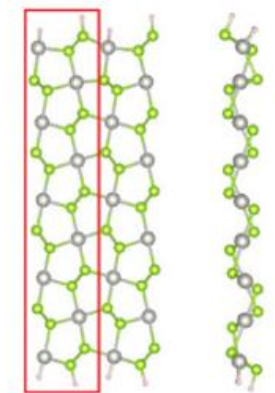
(b)



(f)



(d)



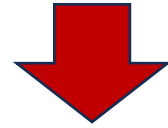
(h)

Hydrogen-passivated pentagonal PdSe₂ nanoribbon optimization models with four distinct edge topologies [(a)AA-3, (b)ZZ-3, (c)ZA-3, (d)SS-2 configurations]

The four edge types [(e)AA, (f)ZZ, (g)ZA, (h)SS, (i)atom legend] discussed in the literature.^[14]

2.4 Simulation calculation of electronic properties

Calculation Basis: Adopt fully relaxed stable structures after different vdW corrections to explore structure-induced electronic property variations.



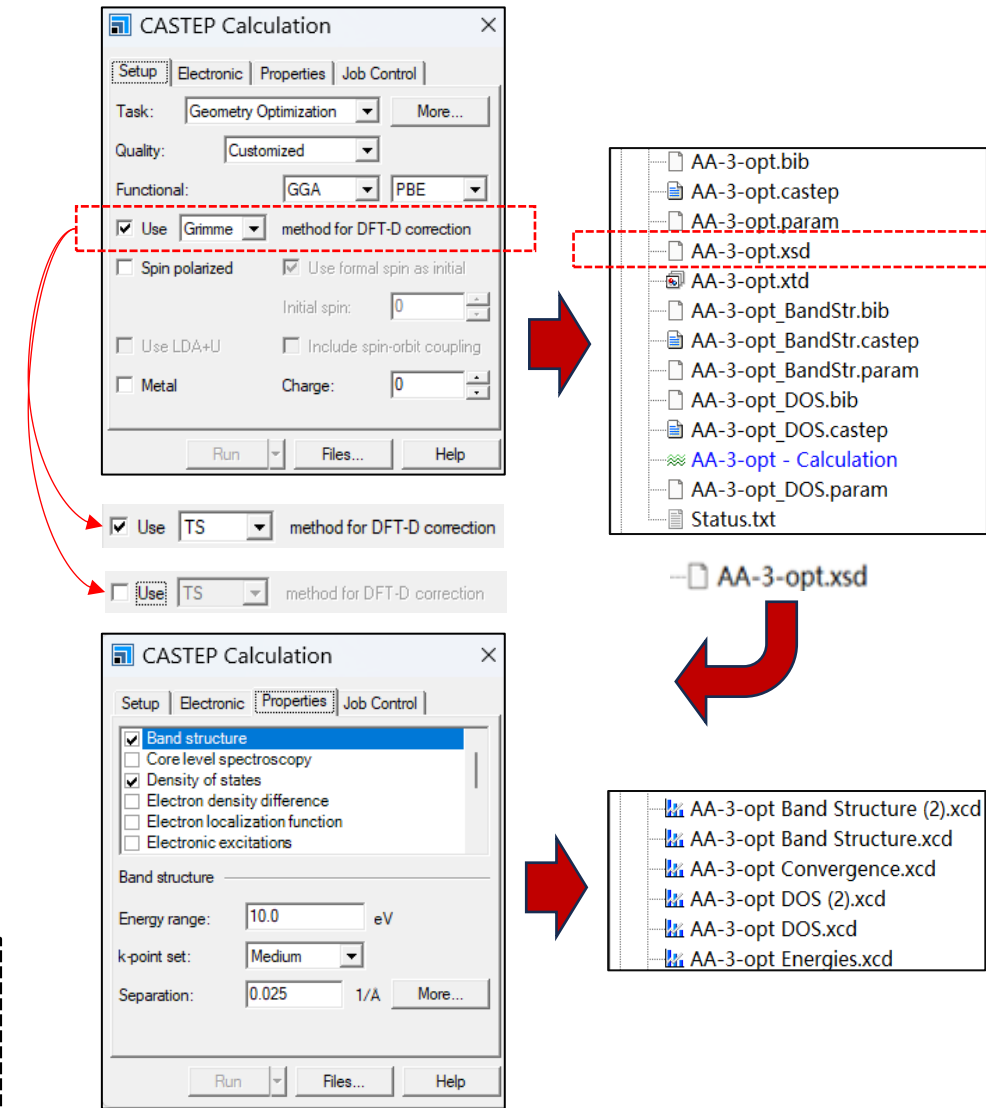
Explore band structure

Band Structure Calculation: Non-self-consistent calculation along high-symmetry paths is adopted to obtain band structure. Band gap **determines electronic category**. Different vdW corrections adjust orbital overlap, tune VBM/CBM dispersion and carrier mobility.



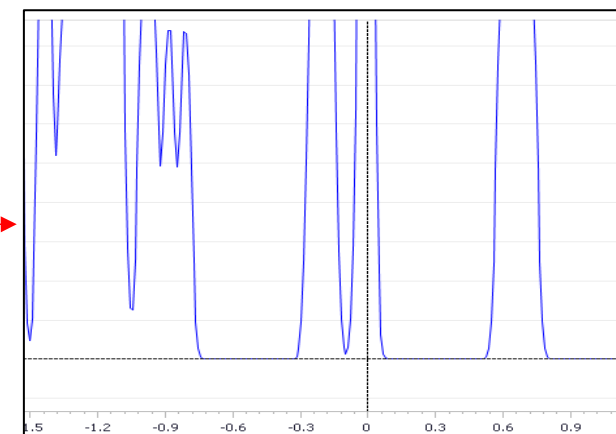
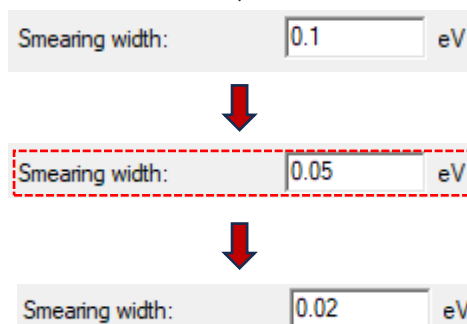
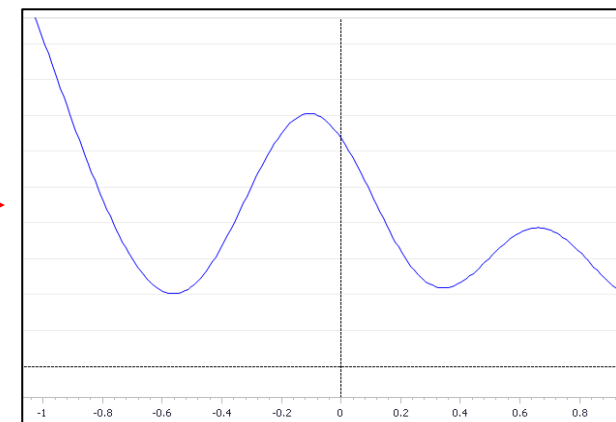
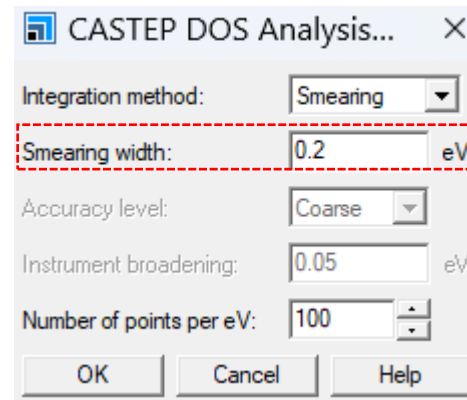
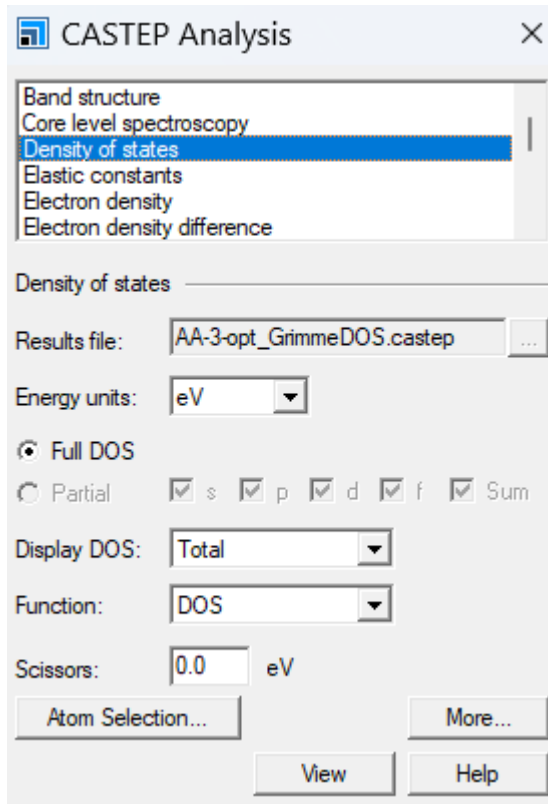
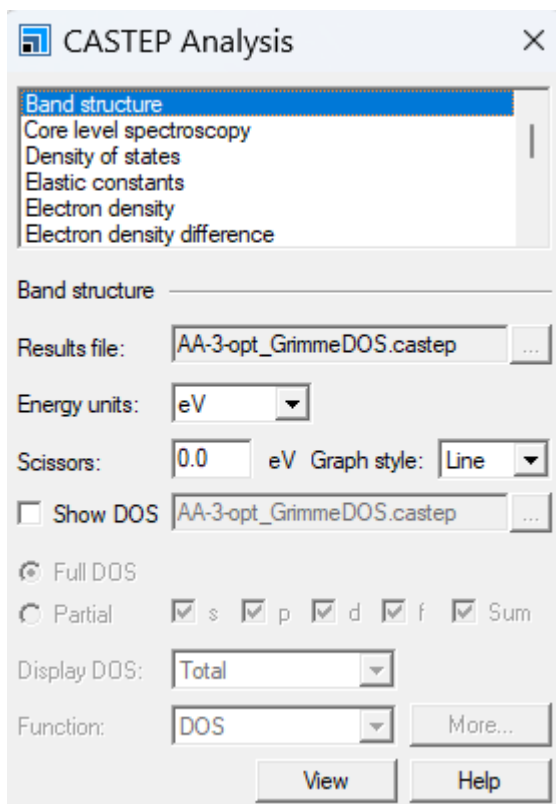
Explore DOS

Density of States Calculation: Total density of states can **judge the electronic conductive properties** of the system. Projected density of states is adopted to **distinguish orbital contributions**, revealing orbital hybridization and energy shift under different van der Waals corrections.



Part of the procedure: AA-3 geometric optimization and electronic properties calculation

Analysis basis: investigate the band gap evolution of PdSe₂ nanoribbons with different configurations (e.g., AA-3: Grimme, TS, PBE)



Part of the procedure: AA-3 band structure and DOS analysis

DOS Analysis between different smearing width(Original figure in Materials Studio)

2.6 Structural stability and geometric reconstruction

Pristine H-passivated nanoribbons exhibit rigid frameworks with negligible structural variation.

Grimme band gap for AA-3: 0.581 eV

PBE band gap for AA-3: 0.587 eV



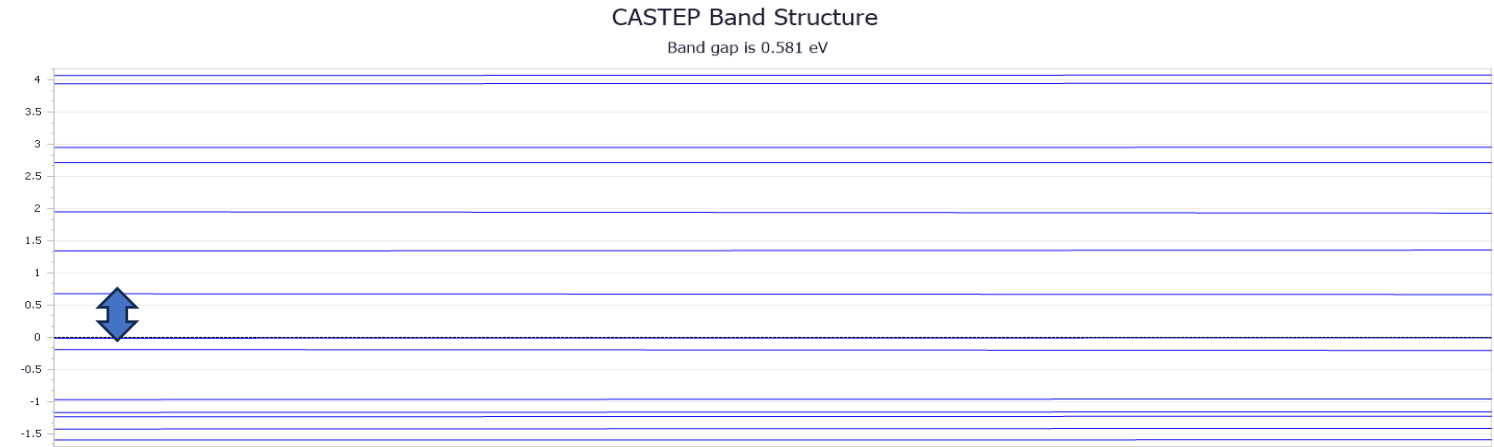
Change the functional groups

Polar functional groups (-F, -OH) significantly change local edge interactions.

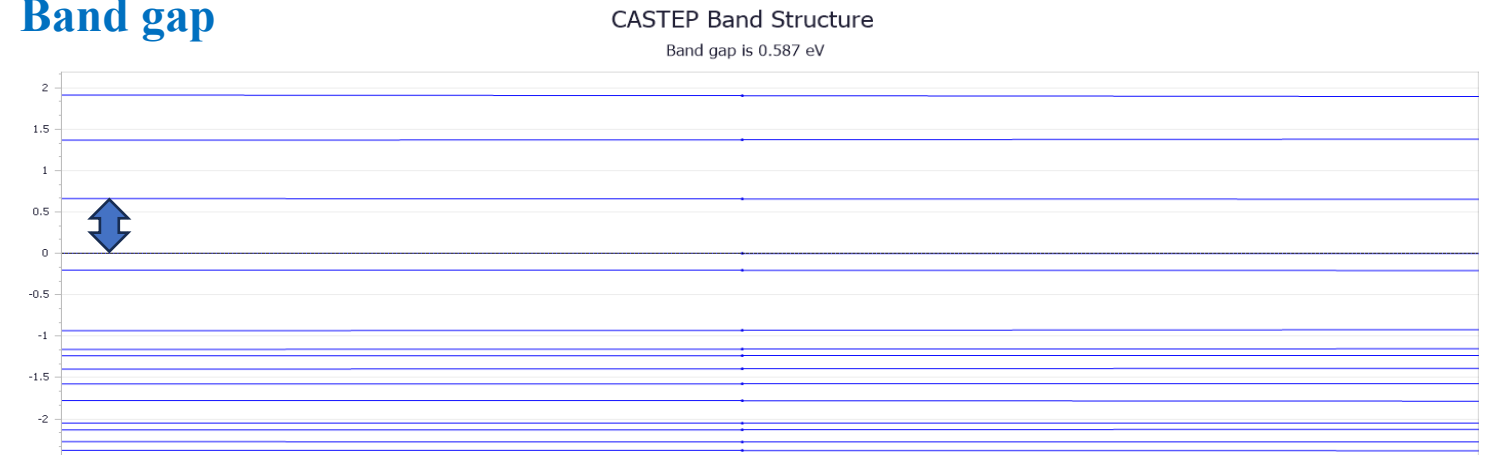


Reconstruction and Reuse methods

Standard PBE neglects long-range attractive forces and overestimates edge atomic distances. After introducing **dispersion correction**, dispersion forces overcome steric hindrance.

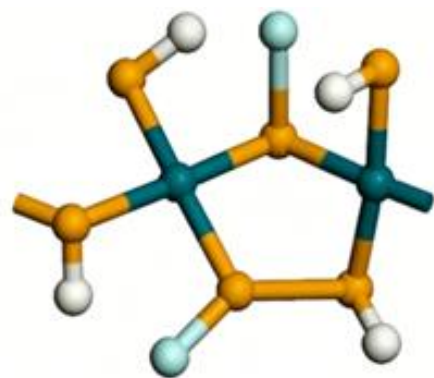


Band gap

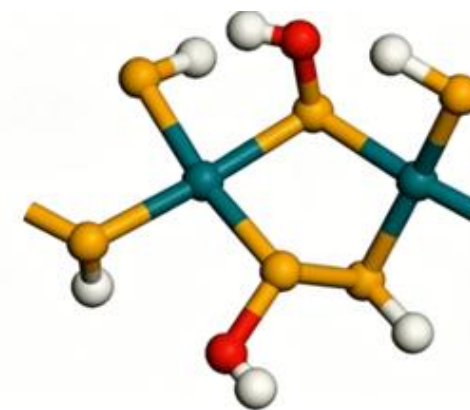
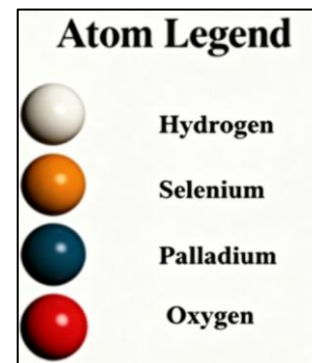
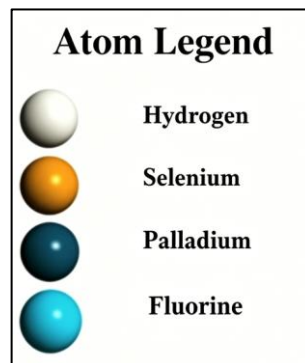


Part of Castep band structure analysis: AA-3 Grimme and AA-3 PBE

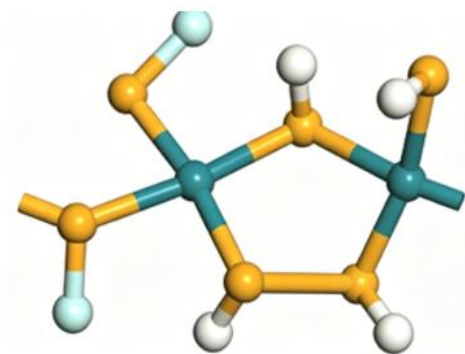
PdSe₂ Edge Chemical Modification of Central and Lateral Substituted



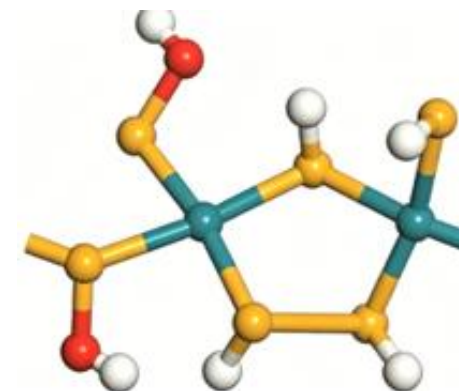
(a)



(b)



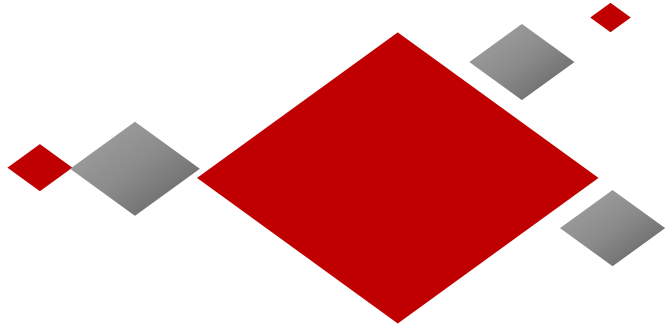
(c)



(d)

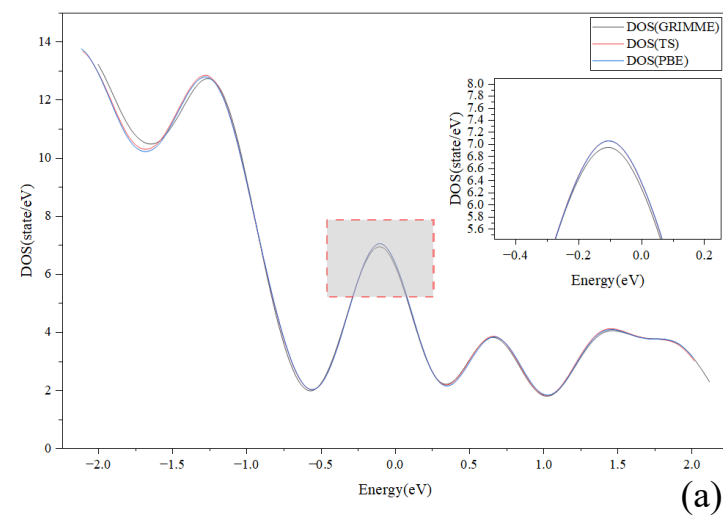
Why choose
Highly electronegative functional groups?
Fluorine and Hydroxyl exhibit
Extreme Electronegativity,
Perfect H-bond Acceptors,
Exposing Functional Flaws
Inducing **electronic delocalization.**

Central (a-b) and Lateral(c-d) Fluorine-substituted or Hydroxyl-substituted pentagonal PdSe₂ nanoribbon optimization

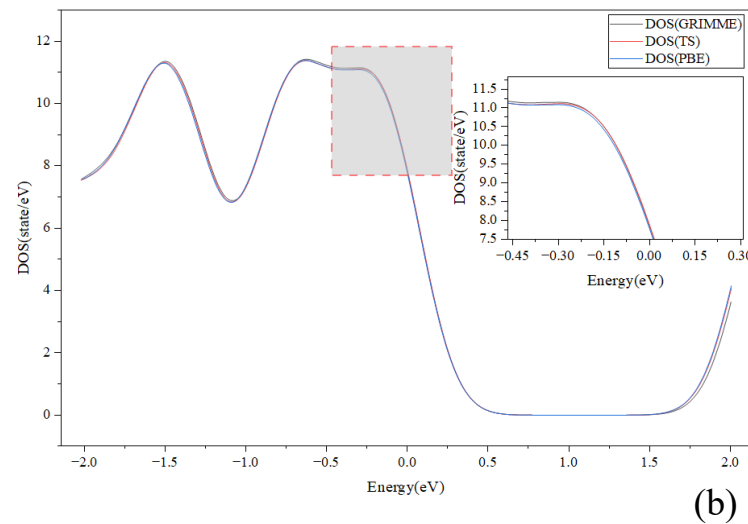


3. Results and discussion

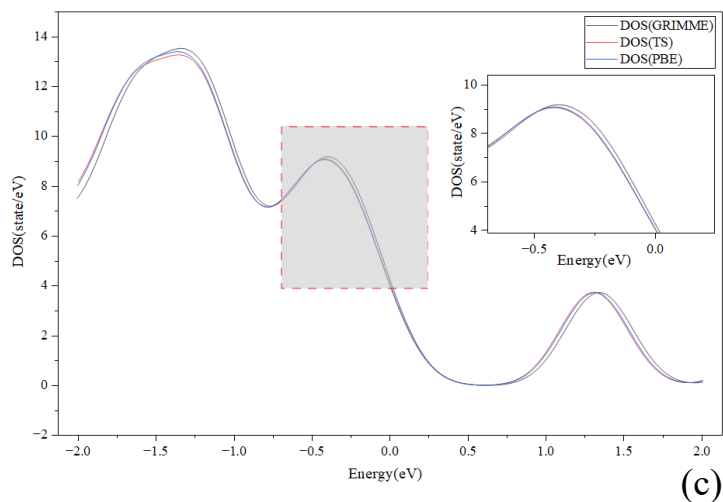




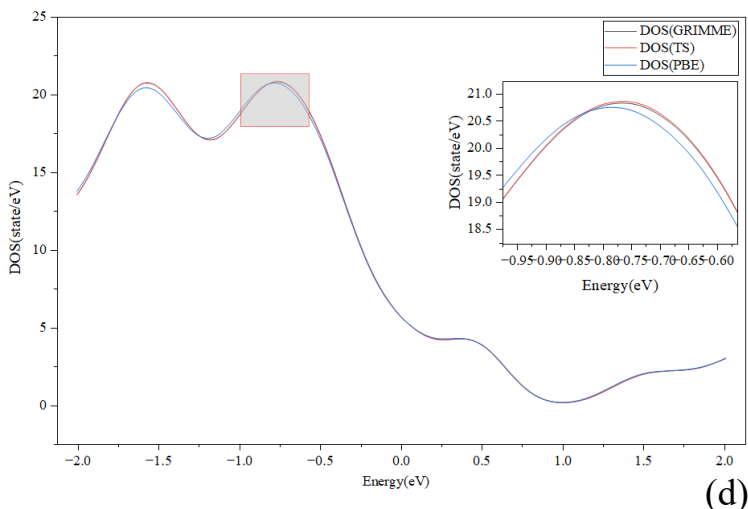
(a)



(b)



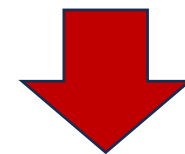
(c)



(d)

Density of states (DOS) comparison of the hydrogen-passivated pentagonal PdSe₂ nanoribbon with (a)AA-3, (b)SS-3, (c)ZA-2, (d)ZZ-3 configuration under different dispersion correction schemes

absolute bandgap deviations of these four configurations are extremely marginal in Chapter 2.6.



Via DOS comparison

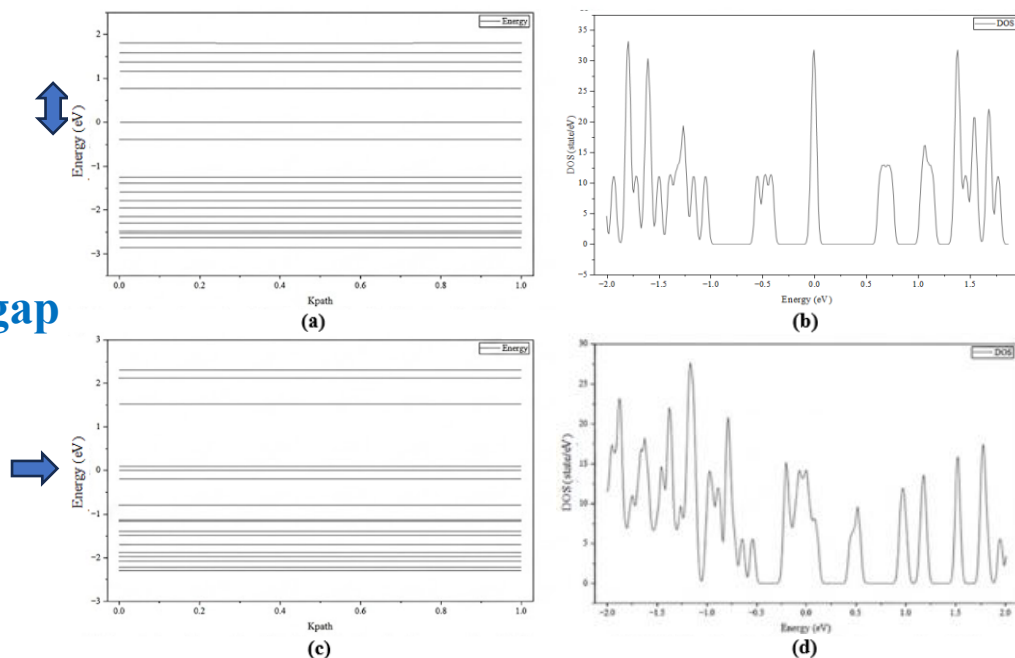
High Consistency in Insets: Only marginal deviations in DOS peak positions and intensities among different functionals.



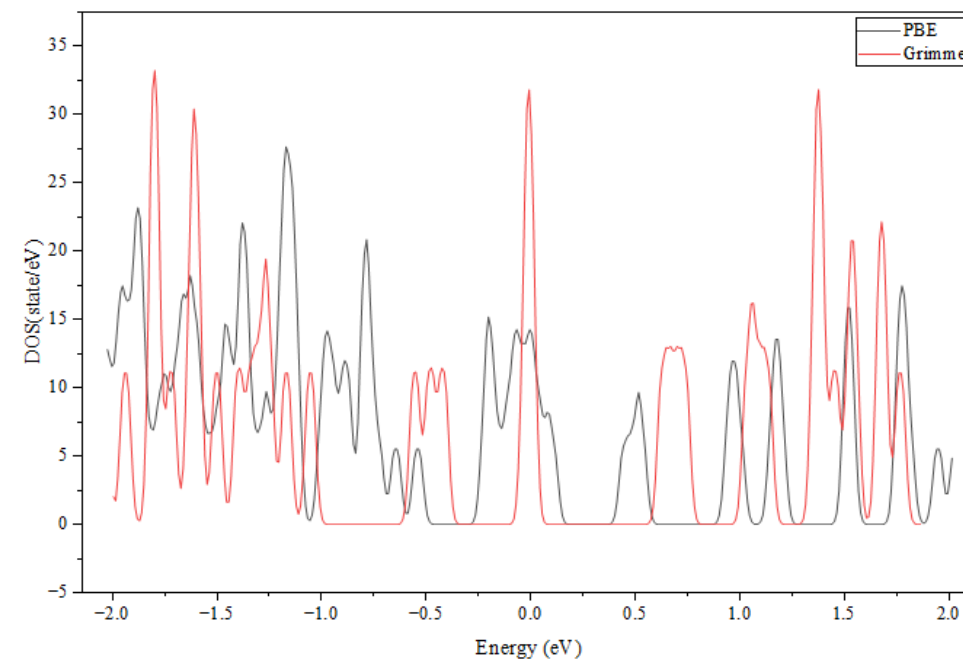
Result 1

Standard PBE functional is fully adequate for describing conventional PdSe₂ nanoribbons without strongly polar functional groups.

Band gap



Band structures and DOS comparisons for the Central F-substitution under Grimme vs. PBE methods. The calculated band structures and DOS using the Grimme correction are presented in (a) and (b), while those using the standard PBE functional are shown in (c) and (d).

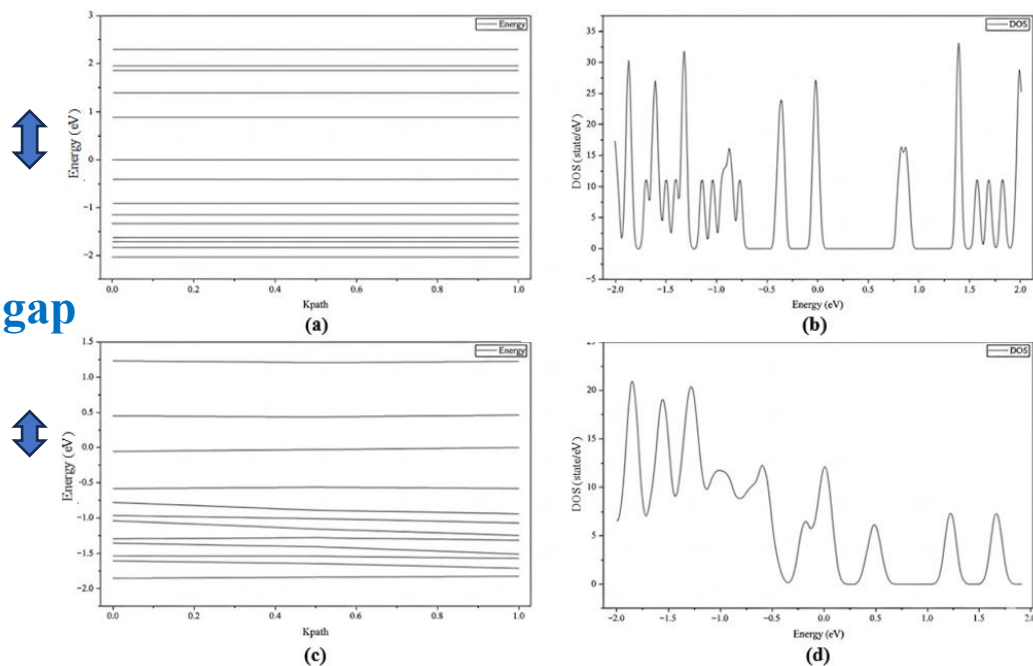


Comparison of DOS between Grimme and PBE methods for AA-3, Fluorine-substituted

Result 2, part 1

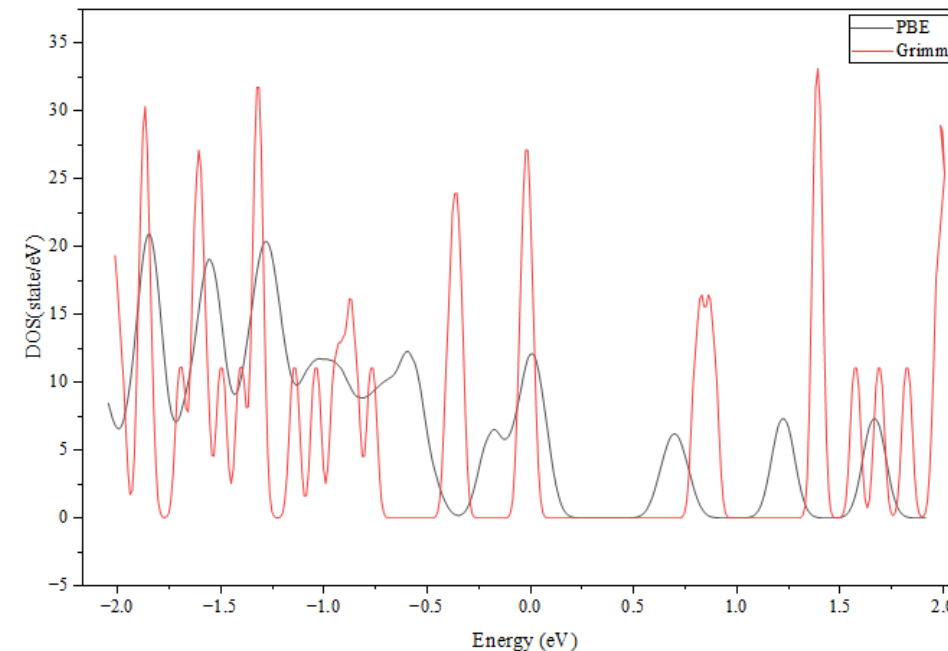
The corrected band gap of central Fluorine substituted AA-3 rises to **0.87 eV**, restoring its stable moderate-band-gap semiconductor property.

Band gap



Band structures and DOS comparisons for the Central OH-substitution under Grimme vs. PBE methods.

The calculated band structures and DOS using the Grimme correction are presented in (a) and (b), while those using the standard PBE functional are shown in (c) and (d).



Comparison of DOS between Grimme and PBE methods for AA-3, Hydroxyl-substituted

Result 2, part 2

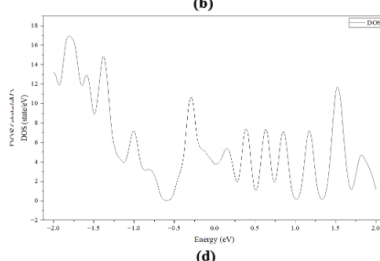
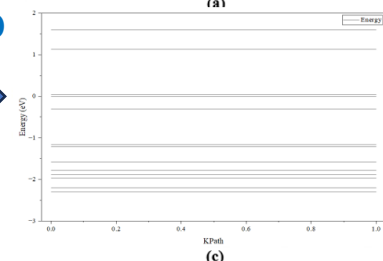
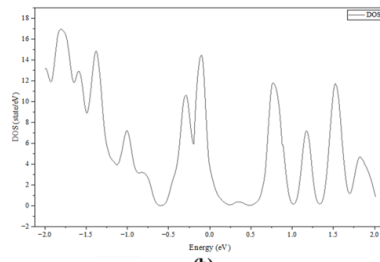
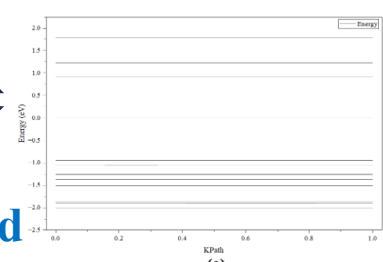
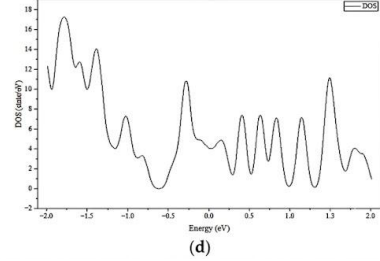
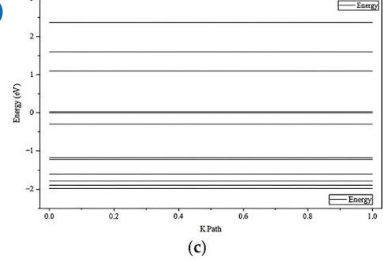
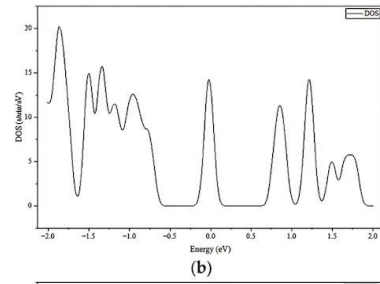
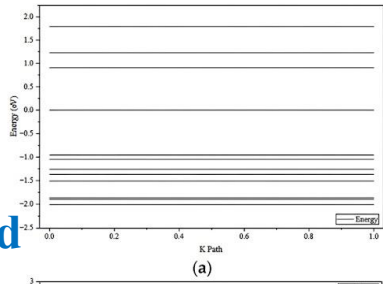
The corrected band gap of central Hydroxyl substituted AA-3 rises to **0.88 eV**, restoring its stable moderate-band-gap semiconductor property.

3.3 Lateral site verification

Band gap



Band gap



Extra Operation

Result 2, part 3

Employ **spin-polarized calculations** to capture localized magnetic moments. **fixed occupancy** is disabled.
The **thermal smearing** factor is increased to 35% to stabilize Fermi distribution.

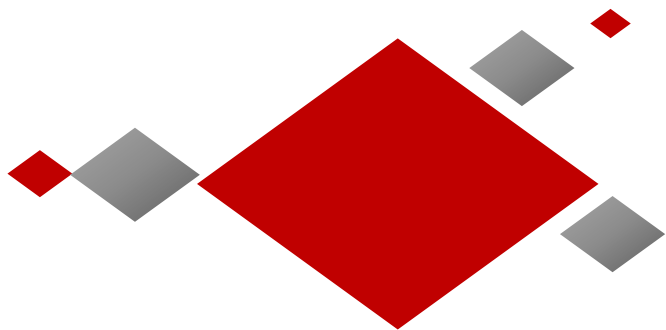
Similarly to part 1&2
Bandgap recovery&reshaping
Corrected F-side: 0.907 eV
Corrected OH-side: 0.87 eV

Summary of Result 2

Functionalized Bandgap

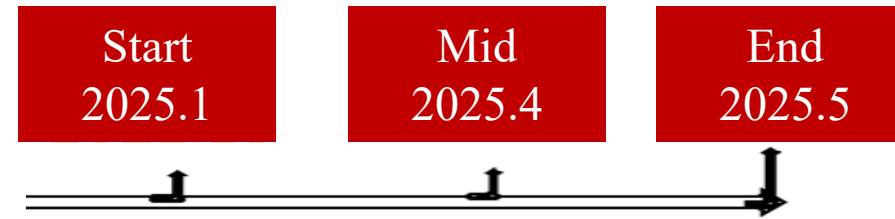
Modification Substituted	PBE	Grimme
Central Fluorine	0.015 eV	0.87 eV
Central Hydroxyl	0.47 eV	0.88 eV
Lateral Fluorine	0.037 eV	0.907 eV
Lateral Hydroxyl	0.048 eV	0.87 eV

Band structures and DOS comparisons for the Central F-substitution and OH-substitution under Grimme vs. PBE methods.
The calculated band structures and DOS using the Grimme correction are presented in (a) and (b), while those using the standard PBE functional are shown in (c) and (d).



4. Conclusion





Conclusions

- i. **H-passivated structures** are determined dispersion effect has little influence on band gap.
- ii. The electronic features of intrinsic by edge topology. Strong covalent framework is rigid, using edge functionalization, which forms local hydrogen bond networks. On F substitution, Standard PBE leads to unphysical electron delocalization, abnormal narrow band gap and artificial metallic states.
- iii. Vertificate with OH substitution, **Dispersion correction** eliminates calculation errors and stabilizes edge structures.

Future Perspectives

- i. Explore **multi-group co-modulation** and external field regulation to realize tunable band gaps.
- ii. Calculate **diverse configurations** like AA-5, AA-7, to clarify the influence of DFT correction more accurately.
- iii. Guide **experimental synthesis** of modified nanoribbons, and verify theoretical results via advanced characterization techniques.



大连理工大学
DALIAN UNIVERSITY OF TECHNOLOGY

Thank you for your attention!
Your comments and suggestions are
highly appreciated.

